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Method for operating a control device for controlling an electric motor, in particular a steering system

Abstract

In a method for operating a control device for controlling an electric motor, in particular a steering system, in which the control device includes at least one power supply, at least one power element which is operatively connected to the power supply, at least one reverse polarity protection switch arranged in a power current path between the power supply and the power element, and at least one logic unit, at least one first voltage value correlated with an intermediate circuit voltage and at least one second voltage value correlated with a supply voltage of the logic unit are determined and linked to one another and evaluated to determine a faulty actuation of the reverse polarity protection switch.

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Background/Summary

(1) This application claims priority under 35 U.S.C. § 119 to application no. DE 10 2022 205 740.6, filed on Jun. 7, 2022 in Germany, the disclosure of which is incorporated herein by reference in its entirety.

BACKGROUND

(2) The disclosure proceeds from a method for operating a control device for controlling an electric motor, in particular a steering system.

(3) In addition, the disclosure relates to a computing unit for carrying out such a method, an actuator assembly comprising such a computing unit, and a steering system comprising such an actuator assembly.

(4) From the prior art, such as DE 10 2019 206 693 A1, control devices for controlling an electric motor, for example in the form of a servomotor of a steering system, are known, wherein a reverse polarity protection switch is used to protect a downstream circuit in the event of a reverse polarity of a supply voltage, by means of which a power current path can be interrupted in such a reverse polarity situation.

(5) When operating such a control device, a basic distinction must be made between the following cases: 1. Normal operation: The reverse polarity protection switch is switched on after the system is switched on. 2. Faulty actuation of the reverse polarity protection switch: The reverse polarity protection switch is not switched on after the system is switched on, for example due to a hardware error. 3. Defect of the reverse polarity protection switch: The reverse polarity protection switch has a fault and is permanently low impedance (“Short”) or permanently high impedance (“Open”).

(6) Within this context, the second case has the highest probability of fault, since several components interact in the control of the reverse polarity protection switch, so that there is a need to reliably diagnose this fault.

(7) Based on this, the task of this disclosure is in particular to provide a simple method for operating a control device for controlling an electric motor with improved properties with regard to fault diagnosis. The problem is solved by the features described herein, while advantageous configurations and further developments can also be found herein.

SUMMARY

(8) The disclosure is based on a method, in particular a computer-implemented method, for operating a control device for controlling an electric motor, in particular of a vehicle and advantageously of a steering system, wherein the control device comprises at least one power supply, at least one power element operatively connected to the power supply, at least one reverse polarity protection switch arranged in a power current path between the power supply and the power element, and at least one logic unit. In particular, the control device can also include the power current path, which electrically connects the power supply and the power element.

(9) It is proposed that at least one first voltage value correlated with an intermediate circuit voltage and at least one second voltage value correlated with a supply voltage of the logic unit be determined, and be linked with each other and evaluated to determine a faulty actuation of the reverse polarity protection switch. The first voltage value is advantageously tapped after the reverse polarity protection switch, i.e. in particular on a side that faces the power element and/or the

electric motor, while the second voltage value is tapped before the reverse polarity protection switch, i.e. in particular on a side that faces the power supply. In particular, the first voltage value in this context can also correspond to the intermediate circuit voltage, while the second voltage value can be identical to the supply voltage of the logic unit. In principle, however, voltage values deviating from this can also be determined, however these are preferably linked to the intermediate circuit voltage or the supply voltage of the logic unit in such a manner that the intermediate circuit voltage or the supply voltage of the logic unit can or could be determined on the basis of the determined voltage values. This configuration can improve fault diagnosis, in particular for faulty actuation of a reverse polarity protection switch. In particular, fault diagnosis can be carried out particularly simply and efficiently within this context, whereby additional components can be advantageously dispensed with and costs minimized.

(10) Within this context, a “control device” is to be understood to refer in particular to at least one part, in particular a subassembly, of an actuator assembly, which is provided in at least one operating state for controlling an operation of at least one electric motor of the actuator assembly. The electric motor is in particular configured as a servo motor, advantageously as a brushless motor, and particularly advantageously as an asynchronous motor or as a permanently excited synchronous motor. Preferably, the electric motor is provided as part of an auxiliary electric power steering and in particular for generating an electric steering assistance. The electric motor can, for example, be designed as a three-phase, six-phase, nine-phase or twelve-phase electric motor. In addition, the power element comprises at least one power electronics unit, which is designed in particular as an inverter unit, in particular as an output stage and/or as a bridge circuit, and is provided for supplying power and/or current to the electric motor. For this the power electronics can comprise several power switches, preferably designed as field-effect transistors, for example in the form of MOSFETs. The power element can furthermore comprise at least one intermediate circuit capacitor or several intermediate circuit capacitors, preferably connected in parallel. Furthermore, a “reverse polarity protection switch” is to be understood in particular as a switching unit which is provided to protect a downstream unit, in particular the power element and/or the electric motor, in a reverse polarity situation. For this purpose, the reverse polarity protection switch has at least one, in particular controllable, control terminal, advantageously a base terminal and/or a gate terminal, at least one, in particular switchable, first power terminal, advantageously an emitter terminal and/or a source terminal, and at least one, in particular switchable, second power terminal, in particular a collector terminal and/or a drain terminal. In addition, the reverse polarity protection switch comprises at least one switching element, advantageously a semiconductor switching element, and preferably a diode, in particular one connected in parallel with the switching element, which can be designed as a body diode, for example. Preferably, the reverse polarity protection switch is herein designed as a transistor and particularly advantageously as a MOSFET. A switching state of the reverse polarity protection switch is further controllable in particular by a control voltage, in particular between the control terminal and the first power terminal, preferably a gate-source voltage. In addition, a “logic unit” is to be understood as an at least partially electrically and/or electronically designed unit, which is in particular electrically connected to the power element, and is preferably provided to perform at least one control functionality, at least one evaluation functionality and/or at least one monitoring functionality. Within this context, the logic unit can be designed as an integrated circuit, preferably as a system base chip, and can comprise, for example, an integrated watchdog unit.

(11) Furthermore, a computing unit can be provided, which is intended to carry out the method for operating the control device. The term “computing unit” is in particular intended to mean an electrical and/or electronic unit which comprises an information input, an information processor, and an information output. Advantageously, the computing unit furthermore comprises at least one processor, at least one memory, at least one input and/or output device, at least one operating program, at least one control routine, at least one calculation routine, at least one evaluation routine

and/or at least one determination routine. In the present case, the computing unit can be provided, in particular by means of the determination routine, for determining at least one first voltage value correlated with an intermediate circuit voltage and/or at least one second voltage value correlated with a supply voltage of the logic unit. Furthermore, the computing unit is provided at least to determine, in particular by means of the evaluation routine, a faulty actuation of the reverse polarity protection switch, and to link at least the first voltage value and the second voltage value with each other for this purpose and evaluate them. The computing unit is in this case preferably integrated into a control device of the vehicle, e.g., a central vehicle control device, or a control device of the steering system, in particular in the form of a steering control device. The term “provided” is in particular intended to mean specifically programmed, designed, and/or equipped. The phrase “an object being provided for a specific function” is in particular intended to mean that the object fulfills and/or performs this specific function in at least one application and/or operating state.

(12) It is furthermore proposed that a difference between the first voltage value and the second voltage value be determined in order to determine the faulty actuation of the reverse polarity protection switch. Particularly advantageously, the difference between the first voltage value and the second voltage value is then compared with a limit value. The limit value is advantageously greater than 2 V and/or less than 20 V. Typically, the limit value can be within the range between 4 V and 10 V, for example. Preferably, it is also proposed that a system response be triggered in the event that the difference between the first voltage value and the second voltage value exceeds the limit value. Within this context, the system response can consist, for example, of an immediate shutdown of the system or a switch to a redundant further system, for example from a first partial machine of the electric motor to a second partial machine of the electric motor. Preferably, however, the system response comprises ramping a motor torque of the electric motor. Alternatively or additionally, it is conceivable to generate an information message. With this configuration, an advantageously simple evaluation algorithm can be provided, and/or a particularly reliable fault diagnosis can be achieved.

(13) According to a preferred configuration, it is proposed that the first voltage value and/or the second voltage value be determined in an operating state in which the direction of rotation of the electric motor changes, for example due to a reversal at a steering hand grip of the steering system. Advantageously, it is also proposed within this context that the electric motor in the operating state change from operation in a first direction, for example in a left-hand direction and/or a forward direction, to operation in a second direction, for example in a right-hand direction and/or a reverse direction, so that an energy recovery takes place which leads at least to an increase in the first voltage value and consequently to an increase in the difference between the first voltage value and the second voltage value. Determining the voltage values during operation of the electric motor in reverse direction or during a change of the electric motor in reverse direction has the advantage that a high voltage difference between the voltage values occurs due to the change of direction, which can be determined by means of a conventional and/or simply constructed measuring device. As a result, it is advantageous to dispense with additional components and/or precise measuring equipment, thereby minimizing costs.

(14) Furthermore, it is proposed that the first voltage value be determined in the region of the power element, preferably between the at least one intermediate circuit capacitor and the power electronics unit, and the second voltage value is determined in the region of the logic unit, preferably directly upstream of a supply terminal of the logic unit, wherein the power element and the logic unit are decoupled from one another by means of a decoupling unit, in particular in such a way that the first voltage value and the second voltage value are also decoupled from one another. In this case, the decoupling unit particularly preferably comprises at least one diode and at least one capacitor. In this manner, an advantageous decoupling between the voltage values for determining the faulty actuation of the reverse polarity protection switch can be achieved. In addition, this

process has the advantage that voltage drops in the power element, which can also occur during normal operation, do not lead to a failure or reset of the logic.

(15) Further simplification of the fault diagnosis can also be achieved if an analogue-to-digital converter is used to determine the first voltage value and/or the second voltage value. Particularly preferably, a first analogue-to-digital converter is used to determine the first voltage value and a second analogue-to-digital converter is used to determine the second voltage value.

(16) In addition, the disclosure relates to an actuator assembly with an electric motor, in particular the electric motor mentioned above, with a control device, in particular the aforementioned control device, and with a computing unit, in particular the aforementioned computing unit, for carrying out the method. Particularly preferably, the control device, the electric motor and the computing unit are part of a steering system, which is intended in particular for use in a vehicle and preferably a motor vehicle.

(17) The method for operating the control device is not intended to be limited to the application and embodiment described above. In particular, the method of operating the control device to perform a function described herein can utilize a different number of individual elements, components and units from a number described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further advantages will become apparent from the description of the drawings hereinafter. The drawings illustrate an embodiment example of the disclosure.

(2) Here:

(3) FIG. 1 a part of an exemplary steering system with an actuator assembly comprising an electric motor, a control device and a computing unit in a perspective view,

(4) FIG. 2 the control device in a schematic principle illustration, and

(5) FIG. 3 an exemplary flow chart with main method steps of a method for operating the control device.

DETAILED DESCRIPTION

(6) The following embodiment example relates, by way of example, to a steering system. However, in principle, the disclosure is not limited to use in a steering system and could also be used in other areas of a vehicle, for example a wiping system, a window regulator system, and/or a drive system, and/or in other electronic systems, for example in the area of household appliances and/or machine tools.

(7) FIG. 1 shows at least a portion of an exemplary steering system **14** in a perspective view. In the present case, the steering system **14** is configured as an electrically assisted steering system. The steering system **14** is configured by way of example as a conventional steering system and comprises an auxiliary electric steering in the form of a power steering. Furthermore, the steering system **14** is provided for use in a vehicle (not shown), in particular a motor vehicle. In an installed state, the steering system **14** is operatively connected to the vehicle wheels (not shown), and is provided in order to influence a direction of travel of the vehicle. Alternatively, however, it is also contemplated to configure a steering system having an electrical superimposed steering and/or active steering. A steering system could furthermore in principle also be configured as a steer-by-wire steering system and/or as a rear axle steering. Moreover, a corresponding steering system could be provided for use in a commercial vehicle.

(8) The steering system **14** comprises a steering gear **44**, exemplarily designed as a rack and pinion steering gear, which is provided to convert a steering input into a steering movement of the vehicle wheels. For this purpose, the steering gear **44** comprises at least one steering adjustment element **46**, which in the present case is designed in particular as a toothed rack.

(9) Furthermore, the steering system **14** comprises at least one actuator assembly **42**. The actuator assembly **42** is designed as a steering actuator and has an operative connection with the steering adjustment element **46**. The actuator assembly **42** is provided to provide steering torque. In the present case, the actuator assembly **42** is intended to provide a steering torque in the form of an assistance torque and/or servo torque and to apply it, in particular for steering assistance, to the steering gear **44**. Alternatively, however, an actuator assembly could also be part of an electrical superimposed steering and/or active steering, and is in particular provided for providing an additional steering angle and/or variable gear ratio. Furthermore, an actuator assembly could be part of a steer-by-wire steering system. In this case, the actuator assembly could be particularly provided for use in a wheel steering angle adjuster and in particular to provide a steering torque for direct control of a direction of travel of a vehicle. In this case, the actuator assembly could also be provided for use in a control unit of the steer-by-wire steering system and for providing a feedback torque and/or resetting torque to a steering handle. Furthermore, as mentioned above, an actuator assembly could also be used independently of a steering system.

(10) The actuator assembly **42** comprises an electric motor **12** that is known in itself. The electric motor **12** is configured as a synchronous motor, in particular a permanently excited motor. The electric motor **12** is further configured as a multi-phase electric motor. In the present case, the electric motor **12** is configured by way of example as a three-phase electric motor. The electric motor **12** is operatively connected to the steering gear **44**, in particular the steering adjustment element **46**. The electric motor **12** is provided in order to generate the steering torque. In the present case, the electric motor **12** is part of the auxiliary electric power steering and is in particular used in order to generate the electric steering assistance. In principle, however, an electric motor could also be configured as a six-phase or a twelve-phase electric motor.

(11) Furthermore, the actuator assembly **42** comprises a control device **10**. In the present case, the control device **10** is designed as a control device, in particular as a steering control device. The control device **10** has an operative connection with the electric motor **12** and is provided for controlling an operation of the electric motor **12**.

(12) In addition, the actuator assembly **42** has a computing unit **40**. The computing unit **40** is integrated into the control device **10**. The computing unit **40** comprises at least one processor (not shown), for example in the form of a microprocessor, and at least one operating memory (not shown). Furthermore, the computing unit **40** comprises at least one operating program stored in the operating memory with at least one calculation routine, at least one control routine, at least one determination routine and at least one evaluation routine. In principle, however, it is also conceivable to design a computing unit separately from a control device.

(13) FIG. 2 shows the control device **10** in detail. The control device **10** comprises a power supply **16**, a power element **18** and at least one power current path **20** which electrically connects the power supply **16** and the power element **18** to one another.

(14) The power supply **16** is herein designed as a vehicle battery. A first terminal of the power supply **16**, in particular a positive voltage terminal, is connected to the power current path **20** and is used in particular to provide a supply voltage. A second terminal of the power supply **16** can be electrically connected to a further power current path and/or a ground terminal. In the present case, the power supply **16** is intended to provide a supply voltage, which can be approximately +12 V, for example. In principle, however, a power supply could also provide a deviating supply voltage.

(15) The power element **18** comprises a power electronics unit **48** that is known in itself. The power electronics unit **48** is designed as an output stage and comprises several half-bridges, wherein one of the half-bridges is assigned to each phase of the electric motor **12**. A first terminal of the power electronics unit **48** is electrically connected to the power current path **20** and serves in particular to receive the supply voltage. A second terminal of the power electronics unit **48** can be electrically connected to the further power current path and/or the ground terminal. Further terminals of the power electronics unit **48** are electrically connected to the electric motor **12** and

serve in particular to supply the electric motor **12**.

(16) Furthermore, the power element **18** has at least one intermediate circuit capacitor **50**, which is provided for coupling the power supply **16** and the power electronics unit **48**. In this example, the power element **18** comprises three intermediate circuit capacitors **50**, which are connected in parallel with each other. A first terminal of the intermediate circuit capacitor **50** or intermediate circuit capacitors **50** is electrically connected to the power current path **20** and/or the first terminal of the power electronics unit **48**. A second terminal of the intermediate circuit capacitor **50** or intermediate circuit capacitors **50** can be electrically connected to the further power current path and/or the ground terminal. The intermediate circuit capacitor **50** or intermediate circuit capacitors **50** are connected in parallel to the power electronics unit **48**.

(17) In addition, the control device **10** can comprise further components arranged in the power current path **20**, such as, for example, an inductor **52**, in particular in the form of a throttle. In principle, however, it is also conceivable to omit such an inductor.

(18) Furthermore, the control device **10** comprises a reverse polarity protection switch **22**. For clarity, the individual components of the reverse polarity protection switch **22** are not marked with reference numbers in FIG. **2**. The reverse polarity protection switch **22** is located in the power current path **20**. In the present case, the reverse polarity protection switch **22** is designed as a MOSFET. The polarity reversal protection switch **22** comprises a control terminal, in particular designed as a gate terminal, a first power terminal, in particular designed as a source terminal, and a second power terminal, in particular designed as a drain terminal. The first power terminal is electrically connected to the inductor **52** and/or the first terminal of the power supply **16**. The second power terminal is electrically connected to the first terminal of the power electronics unit **48**. In addition, the second power terminal is electrically connected to the first terminal of the intermediate circuit capacitor **50** or intermediate circuit capacitors **50**. Furthermore, the reverse polarity protection switch **22** comprises a body diode. The body diode is oriented such that an anode terminal of the body diode is oriented towards the power supply **16** and a cathode terminal of the body diode is oriented towards the power electronics unit **48** and/or the intermediate circuit capacitor **50** or intermediate circuit capacitors **50**. In principle, however, a reverse polarity protection switch could also be designed as a switch other than a MOSFET, preferably a semiconductor switch.

(19) For controlling the reverse polarity protection switch **22**, the control device **10** comprises a driver unit **54** that is known in itself. In terms of circuitry, the driver unit **54** is arranged between the control terminal of the reverse polarity protection switch **22** and the first power terminal of the reverse polarity protection switch **22**. The driver unit **54** is provided to provide a control voltage for the reverse polarity protection switch **22**.

(20) Furthermore, the control device **10** comprises a logic unit **24**. The logic unit **24** is designed as an integrated circuit, in particular as a system base chip, and comprises an integrated watchdog unit (not shown). The logic unit **24**, in particular a supply terminal of the logic unit **24**, is electrically connected to the power current path **20** and is arranged in particular on a side of the reverse polarity protection switch **22** that faces the power supply **16**. In the present case, the logic unit **24** is arranged between the power supply **16** and the reverse polarity protection switch **22** in terms of circuitry. Here, the logic unit **24** is electrically arranged between the inductor **52** and the first power terminal of the reverse polarity protection switch **22**.

(21) For decoupling the logic unit **24**, the control device **10** can further comprise a decoupling unit **30**, in particular optionally. In terms of circuitry, the decoupling unit **30** is arranged between the power supply **16** and the logic unit **24**. By means of the decoupling unit **30**, the power element **18** and the logic unit **24** are decoupled from each other, so that voltage drops in the power element **18** do not lead to a failure or reset of the logic unit **24**. Herein the decoupling unit **30** comprises at least one diode **32** and at least one capacitor **34** designed as a buffer capacitor. The diode **32** is oriented such that an anode terminal of the diode **32** is oriented towards the power current path **20** and a

cathode terminal of the diode **32** is oriented towards the logic unit **24**. Furthermore, a first terminal of the capacitor **34** is electrically connected to the cathode terminal of the diode **32** and/or the logic unit **24**, in particular the supply terminal of the logic unit **24**, while a second terminal of the capacitor **34** is connected to the ground terminal. In principle, however, a decoupling unit could also have a different capacitance and/or inductance from a capacitor. In addition, instead of one diode, at least two antiseriably connected semiconductor switches or similar could be used. It is furthermore conceivable to entirely dispense with a decoupling unit.

(22) When operating the control device **10**, the following cases are to be distinguished in principle: 1. Normal operation: The reverse polarity protection switch **22** is switched on after the system is switched on. 2. Faulty actuation of the reverse polarity protection switch **22**: The reverse polarity protection switch **22** is not switched on after the system has been switched on, for example due to a hardware error. 3. Defect of the reverse polarity protection switch **22**: The reverse polarity protection switch **22** has a fault and is permanently low impedance (“short”) or permanently high impedance (“open”).

(23) In this context, the second case has the highest probability of failure, since several components fundamentally interact when controlling the reverse polarity protection switch **22**.

(24) Therefore, in order to improve fault diagnosis, a method for operating the control device **10** is proposed below. In particular, the computing unit **40** is provided to carry out the method and has a computer program with corresponding program code means for this purpose. However, another computing unit of a vehicle could alternatively be provided for performing the method.

(25) The computing unit **40** is herein provided for determining at least one first voltage value **26** correlated with an intermediate circuit voltage and at least one second voltage value **28** correlated with a supply voltage of the logic unit **24**, and for linking and evaluating them with one another in order to determine faulty actuation of the reverse polarity protection switch **22**.

(26) For this purpose, the computing unit **40** is electrically connected to the power element **18**, in particular the power current path **20**, by means of a first terminal and in particular via a first analogue-to-digital converter **36**, in particular in such a manner that the computing unit **40** is arranged between the intermediate circuit capacitor **50** or the intermediate circuit capacitors **50** and the power electronics unit **48** in terms of circuitry. In addition, the computing unit **40** is electrically connected to the logic unit **24**, in particular the supply terminal of the logic unit **24**, by means of a second terminal and in particular via a second analogue-to-digital converter **38**, in particular in such a manner that the computing unit **40** is arranged between the decoupling unit **30** and the logic unit **24** in terms of circuitry.

(27) Thus, at least one first voltage value **26** correlated with an intermediate circuit voltage and at least one second voltage value **28** correlated with a supply voltage of the logic unit **24** are read in via a respective analogue-to-digital converter **36**, **38**. The first voltage value **26** is tapped after the reverse polarity protection switch **22**, i.e. on a side that faces the power element **18** and/or the electric motor **12**, while the second voltage value **28** is tapped before the reverse polarity protection switch **22**, i.e. on a side that faces the power supply **16**. The first voltage value **26** is determined in the area of the power unit **18**, in particular between the intermediate circuit capacitor **50** or the intermediate circuit capacitors **50** and the power electronics unit **48**, and the second voltage value **28** is determined in the area of the logic unit **24**, in particular directly before the supply terminal of the logic unit **24**. Subsequently, a difference between the first voltage value **26** and the second voltage value **28** is determined and compared with a limit value in order to determine the faulty actuation of the reverse polarity protection switch **22**. The first voltage value **26** and the second voltage value **28** are determined in an operating state in which the direction of rotation of the electric motor **12** changes, wherein the electric motor **12** changes from operation in a first direction, for example counterclockwise rotation, to operation in a second direction, for example clockwise rotation, so that an energy recovery takes place which leads to at least an increase in the first voltage value **26** and consequently an increase in the difference between the first voltage value **26**

and the second voltage value **28**. In the event that the difference between the first voltage value **26** and the second voltage value **28** exceeds the limit value, for example 4 V, a system response is triggered, wherein the system response can comprise, for example, ramping a motor torque of the electric motor **12**. Accordingly, the present method is provided for determining faulty actuation of the reverse polarity protection switch **22**, and does not serve to determine a fault of the reverse polarity protection switch **22** itself.

(28) The present method is based on the following consideration in order to detect faulty actuation of the reverse polarity protection switch **22**.

(29) In said fault case, the reverse polarity protection switch **22** acts as a diode, since the body diode of the reverse polarity protection switch **22** is still effective. This means that current from the power supply **16** is allowed to pass in the direction of the electric motor **12** (forward operation), while current from the electric motor **12** is blocked in the direction of the power supply **16** (energetic recovery). Energetic recovery occurs, among other things, when the direction of rotation of the electric motor **12** changes, for example simultaneously with the change of direction at a steering hand grip in the form of a steering wheel of the steering system **14**. Such energetic recovery results in the current charging the intermediate circuit capacitor **50** or the intermediate circuit capacitors **50** to a higher voltage, and the first voltage value **26** increases. From a system point of view, this can then be diagnosed by exceeding the limit value. A determination of the voltage values **26**, **28** during energetic recovery, for example caused by a reversal of the direction of rotation of the electric motor **12**, has the advantage that a high voltage difference between the voltage values **26**, **28** occurs due to the change of direction, which is significantly higher than a voltage difference in normal operation, so that conventional and/or simply constructed measuring devices, for example in the form of a voltage divider and an additional analogue-digital converter, can be used.

(30) Finally, FIG. 3 shows an exemplary flow chart with main method steps of the method for operating the control device **10**.

(31) In a method step **60**, at least one first voltage value **26** correlated with an intermediate circuit voltage and at least one second voltage value **28** correlated with a supply voltage of the logic unit **24** are read in via the respective analogue-to-digital converter **36**, **38**.

(32) In a method step **62**, the first voltage value **26** and the second voltage value **28** are linked and evaluated to determine faulty actuation of the reverse polarity protection switch **22**. Within this context, a difference between the first voltage value **26** and the second voltage value **28** is preferably determined and adjusted with a limit value, for example 4 V.

(33) In the event that the difference between the first voltage value **26** and the second voltage value **28** does not exceed the limit value, method step **60** follows again.

(34) In the event that the difference between the first voltage value **26** and the second voltage value **28** exceeds the limit value, a method step **64** follows.

(35) In method step **64**, a system response is initiated, wherein the system response can comprise, for example, ramping a motor torque of the electric motor **12**. Alternatively or additionally, it is conceivable to generate an information message.

(36) The exemplary flow chart in FIG. 3 is only intended to describe an exemplary method for operating the control device **10**. Individual method steps can in particular also vary, or additional method steps can be added. Within this context, for example, it is conceivable to dispense with ramping a motor torque of the electric motor **12**, instead switching off the system immediately. Furthermore, the system response could be to switch to a redundant further system, for example from a first sub-machine of the electric motor **12** to a second sub-machine of the electric motor **12**.

Claims

1. A method for operating a control device for controlling an electric motor, the control device including at least one power supply, at least one power element which is operatively connected to the power supply, at least one polarity reversal protection switch which is arranged in a power current path between the power supply and the power element, and at least one logic unit, the method comprising: determining at least one first voltage value correlated with an intermediate circuit voltage and at least one second voltage value correlated with a supply voltage of the logic unit; and linking the at least one first voltage value and the at least one second voltage value to one another and evaluating the at least one first voltage value and the at least one second voltage value to determine faulty actuation of the reverse polarity protection switch, wherein the at least one first voltage value is determined in a region of the at least one power element and the at least one second voltage value is determined in a region of the at least one logic unit, and wherein the at least one power element and the at least one logic unit are decoupled from one another by a decoupling unit.
2. The method according to claim 1, wherein the determining of the faulty actuation of the reverse polarity protection switch includes: determining a difference between the at least one first voltage value and the at least one second voltage value; and comparing the difference with a limit value.
3. The method according to claim 2, further comprising: triggering a system response in response to the difference between the at least one first voltage value and the at least one second voltage value exceeds the limit value.
4. The method according to claim 3, wherein the system response includes ramping a motor torque of the electric motor.
5. The method according to claim 1, wherein the at least one first voltage value and the at least one second voltage value are determined in an operating state in which a direction of rotation of the electric motor changes.
6. The method according to claim 5, wherein, in the operating state, the direction of the electric motor changes from operation in a first direction to operation in a second direction and an energy recovery takes place resulting in at least an increase in the at least one first voltage value.
7. The method according to claim 1, wherein the decoupling unit includes at least one diode and at least one capacitor.
8. The method according to claim 1, wherein the determining of the at least one first voltage value and the at least one second voltage value includes using an analog-to-digital converter used to determine the at least one first voltage value and/or the at least one second voltage value.
9. The method according to claim 1, wherein the electric motor is included in a steering system.
10. A computing unit comprising: at least one processor; and at least one non-transitory memory comprising stored program instructions that, when executed by the at least one processor, cause the at least one processor to execute the method according to claim 1.
11. An actuator assembly, comprising: an electric motor; a control device including at least one power supply, at least one power element which is operatively connected to the power supply, at least one polarity reversal protection switch which is arranged in a power current path between the power supply and the power element, and at least one logic unit; and a computing unit comprising at least one processor and at least one non-transitory memory comprising stored program instructions that, when executed by the at least one processor, cause the at least one processor to: determine at least one first voltage value correlated with an intermediate circuit voltage and at least one second voltage value correlated with a supply voltage of the logic unit; and link the at least one first voltage value and the at least one second voltage value to one another and evaluating the at least one first voltage value and the at least one second voltage value to determine faulty actuation of the reverse polarity protection switch, wherein the at least one first voltage value and the at least one second voltage value are determined in an operating state in which a direction of rotation of the electric motor changes, and wherein, in the operating state, the direction of the electric motor

- changes from operation in a first direction to operation in a second direction and an energy recovery takes place resulting in at least an increase in the at least one first voltage value.
12. The actuator assembly according to claim 11, wherein the determining of the faulty actuation of the reverse polarity protection switch by the computing unit includes: determining a difference between the at least one first voltage value and the at least one second voltage value; and comparing the difference with a limit value.
13. The actuator assembly according to claim 12, further comprising: triggering a system response in response to the difference between the at least one first voltage value and the at least one second voltage value exceeds the limit value.
14. The actuator assembly according to claim 13, wherein the system response includes ramping a motor torque of the electric motor.
15. The actuator assembly according to claim 11, wherein: the at least one first voltage value is determined in a region of the at least one power element and the at least one second voltage value is determined in a region of the at least one logic unit, and the at least one power element and the at least one logic unit are decoupled from one another by a decoupling unit.
16. The actuator assembly according to claim 15, wherein the decoupling unit includes at least one diode and at least one capacitor.
17. The actuator assembly according to claim 11, wherein the determining of the at least one first voltage value and the at least one second voltage value includes using an analog-to-digital converter used to determine the at least one first voltage value and/or the at least one second voltage value.
18. A steering system comprising: an actuator assembly comprising: an electric motor; a control device including at least one power supply, at least one power element which is operatively connected to the power supply, at least one polarity reversal protection switch which is arranged in a power current path between the power supply and the power element, and at least one logic unit; and a computing unit comprising at least one processor and at least one non-transitory memory comprising stored program instructions that, when executed by the at least one processor, cause the at least one processor to: determine at least one first voltage value correlated with an intermediate circuit voltage and at least one second voltage value correlated with a supply voltage of the logic unit; and link the at least one first voltage value and the at least one second voltage value to one another and evaluating the at least one first voltage value and the at least one second voltage value to determine faulty actuation of the reverse polarity protection switch, wherein the at least one first voltage value is determined in a region of the at least one power element and the at least one second voltage value is determined in a region of the at least one logic unit, and wherein the at least one power element and the at least one logic unit are decoupled from one another by a decoupling unit.
19. The steering system according to claim 18, wherein the decoupling unit includes at least one diode and at least one capacitor.
20. The steering system according to claim 18, wherein: the at least one first voltage value and the at least one second voltage value are determined in an operating state in which a direction of rotation of the electric motor changes, and in the operating state, the direction of the electric motor changes from operation in a first direction to operation in a second direction and an energy recovery takes place resulting in at least an increase in the at least one first voltage value.
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